

SVD: $A = \boxed{} = \boxed{U} \boxed{\Sigma} \boxed{V^T}$

$\nearrow \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n)$

orthogonal

Matrix norms:

- spectral norm: $\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_1$
- Frobenius norm: $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2}$

Low-rank approximation of matrices:

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \rightsquigarrow \text{rank } 2$

want to solve $\min_{\substack{B \in \mathbb{R}^{n \times n} \\ \text{rank}(B)=1}} \|A - B\|_F$

$\rightsquigarrow \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \begin{bmatrix} 1.5 & 2 \\ 3 & 4 \end{bmatrix}, \dots$

Theorem (Eckart-Young theorem)

If $A = U \Sigma V^T = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T$ then a minimizer of $\min_{B, \text{rank}(B) \leq k} \|A - B\|$ is given by the "truncated SVD":

$A_k := \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_k u_k v_k^T$

holds for both the spectral and Frobenius norm

$A = \boxed{} = \boxed{U} \boxed{\Sigma} \boxed{V^T}$

$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$

$\begin{bmatrix} u_1 & \dots & u_k \end{bmatrix} \quad \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{bmatrix} \quad \begin{bmatrix} v_1^T \\ \vdots \\ v_k^T \end{bmatrix}$

$A_k \in \arg\min_{\text{rank}(B) \leq k} \|A - B\|$

$A_k = \boxed{} \boxed{} \boxed{}$

Previous example $\rightsquigarrow A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Example:

	Test A	Test B	Test C
Student 1	90	95	85
Student 2	32	35	31
Student 3	70	75	71
Student 4	70	31	72

$=: A$

(1) If A is rank-1 this means that $A = xy^T = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} [y_1 \ y_2 \ y_3] = \begin{bmatrix} x_1 y_1 & x_1 y_2 & x_1 y_3 \\ x_2 y_1 & x_2 y_2 & x_2 y_3 \\ x_3 y_1 & x_3 y_2 & x_3 y_3 \\ x_4 y_1 & x_4 y_2 & x_4 y_3 \end{bmatrix}$

the score for each student and test is $a_{ij} = x_i \cdot y_j$

$\begin{bmatrix} 5 \\ 10 \\ 2 \\ 7 \end{bmatrix} [100 \ 20 \ 70]$

Annotations: "strength" of the student (pointing to x), difficulty of the test (pointing to y), hardest test! (pointing to 20).

(2) If A is rank-2 this means that $A = xy^T + wz^T$ $x, w \in \mathbb{R}^4$ $y, z \in \mathbb{R}^3$

$a_{ij} = x_i y_j + w_i z_j$

Annotations: how strong the students are in subject 2 (pointing to wz^T), how difficult are the questions about subject 1 (pointing to xy^T), how strong the students are in subject 1 (pointing to xy^T).

If $\min_{\text{rank}(B) \leq 1} \|A - B\|$ is small $\leadsto$ model (1) explains the situation

If not, but $\min_{\text{rank}(B) \leq 2} \|A - B\|$ is small $\leadsto$ model (2) explains the situation.

Q: $A = UZV^T$ $A_k = \begin{bmatrix} u_1 & \dots & u_k \end{bmatrix} \begin{bmatrix} \sigma_1 & \dots & \sigma_k \end{bmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_k^T \end{bmatrix}$

(1) what is an SVD (thin) of $A - A_k$?

(2) what is $\|A - A_k\|_2$?

(3) what is $\|A - A_k\|_F$?

$$A = \sigma_1 u_1 v_1^T + \dots + \sigma_k u_k v_k^T + \sigma_{k+1} u_{k+1} v_{k+1}^T + \dots + \sigma_n u_n v_n^T$$

$$A_k = \sigma_1 u_1 v_1^T + \dots + \sigma_k u_k v_k^T$$

$$A - A_k = \sigma_{k+1} u_{k+1} v_{k+1}^T + \dots + \sigma_n u_n v_n^T = \begin{bmatrix} | & \dots & | \\ u_{k+1} & \dots & u_n \\ | & \dots & | \end{bmatrix} \begin{bmatrix} \sigma_{k+1} & \dots & \sigma_n \end{bmatrix} \begin{bmatrix} -v_{k+1}^T \\ \vdots \\ -v_n^T \end{bmatrix}$$

$$\|A - A_k\|_2 = \sigma_{k+1}$$

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_n^2}$$

Problems

① $\underbrace{A^{-1}(A-B)} \underbrace{B^{-1}(A-B)} = ?$

② Find the inverse of $\begin{bmatrix} A & 0 \\ B & C \end{bmatrix}$ (think $A, B, C \in \mathbb{R}$, then matrices!)

③ Can U orthogonal have an entry > 1 ? (why?)

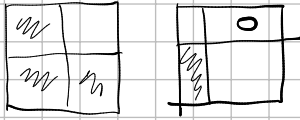
④ When is a diagonal matrix orthogonal? And a triangular (upper) matrix?

⑤ Show that, if λ is an eigenvalue of U orthogonal, then $|\lambda| = 1$

① $(I - A^{-1}B)(B^{-1}A - I) = B^{-1}A - I - \underbrace{A^{-1}B B^{-1}A} + A^{-1}B = B^{-1}A + A^{-1}B - 2I$

② $\begin{bmatrix} A & 0 \\ B & C \end{bmatrix} \begin{bmatrix} x & 0 \\ y & z \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$

$$\begin{cases} AX = I \\ BX + CY = 0 \\ CZ = I \end{cases} \rightarrow \begin{aligned} X &= A^{-1} \\ &\rightarrow BA^{-1} + CY = 0 \rightarrow CY = -BA^{-1} \xrightarrow{\text{multiply with } C^{-1} \text{ to the left}} Y = -C^{-1}BA^{-1} \\ Z &= C^{-1} \end{aligned}$$



③ No! If U orthogonal, all the columns have norm 1 $\Rightarrow$ all entries have $|u_{ij}| \leq 1$.

④ Diagonal matrix orthogonal $\Leftrightarrow \text{diag}(\pm 1, \pm 1, \dots, \pm 1)$

Triangular matrix orthogonal $\Leftrightarrow$ diagonal orthogonal...

$$\begin{bmatrix} u_{11} & & \\ u_{21} & u_{22} & \\ & & \ddots \end{bmatrix} \begin{bmatrix} u_{11}u_{21} & & \\ 0 & u_{22} & \\ & & \ddots \end{bmatrix} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & \ddots \end{bmatrix} \quad u_{11}^2 = 1$$

⑤ λ eigenvalue of U orthogonal $\Rightarrow |\lambda| = 1$.

$$\begin{aligned} Uv &= \lambda v \\ v &\neq 0 \end{aligned} \rightarrow \underbrace{U^T U}_I v = \lambda \cdot U^T v$$

$$v = \lambda \cdot U^T v$$

$$\Rightarrow \|v\|_2 = |\lambda| \cdot \|U^T v\|_2 = |\lambda| \cdot \|v\|_2$$

$$\Rightarrow |\lambda| = 1$$